

A Multimodal Deep Learning Framework for Breast Cancer Receptor Status Prediction

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Abstract. Accurate classification of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2) is essential for breast cancer prognosis and treatment selection, yet standard immunohistochemistry and in situ hybridization (IHC/ISH) can be affected by pre-analytical variability. To address these challenges, we propose a multimodal, multitask deep learning framework that integrates clinical data, hematoxylin and eosin (H&E) whole-slide images (WSIs), and tumor gene-expression profiles, fusing modality-specific encoders through modality-level self-attention for joint ER/PR/HER2 prediction. Developed on 937 matched TCGA-BRCA cases with stratified five-fold cross-validation, the model attained a macro area under the curve (AUC) of 0.890 (ER 0.954, PR 0.868, HER2 0.848), with comparable performance on an independent CPTAC-BRCA external cohort evaluated under a frozen-model protocol with no retraining or tuning (macro AUC 0.931). Ablation showed that multimodal fusion improves over the strongest single modality, and interpretability analyses recovered receptor-associated genes (e.g., ESR1, PGR, ERBB2) and highlighted diagnostically relevant tumor regions. These results suggest multimodal fusion can support interpretable receptor-status prediction as a complementary decision-support tool alongside conventional IHC/ISH.

Keywords: Breast cancer · Receptor status prediction · Multimodal deep learning · Whole Slide Image · Gene expression · Clinical decision support

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1 Introduction

Breast cancer is the most commonly diagnosed cancer in women worldwide and a leading cause of cancer-related mortality [1]. The status of the hormone receptors (ER, estrogen receptor; PR, progesterone receptor) and of human epidermal growth factor receptor 2 (HER2), each scored positive (+) or negative (−), guides prognosis and the choice of therapy (e.g., endocrine therapy, chemotherapy, radiation, surgery) together with factors such as tumor size and stage [2,3].

In standard practice, ER, PR, and HER2 status are assessed by immunohistochemistry (IHC), with in situ hybridization (ISH) for equivocal HER2 cases [2,3]. Although these assays are the clinical gold standard, their reliability can be affected by pre-analytical and analytical factors—cold ischemia time, fixation, staining quality, and observer subjectivity [4]—alongside practical constraints such as turnaround time and cost.

Receptor status manifests as complementary signals across scales—tumor gene expression (molecular), H&E histomorphology (tissue), and clinical context (patient) [5–8]—yet no single modality captures this complexity in isolation. We therefore formulate receptor-status prediction as a supervised problem with three binary targets (ER, PR, HER2).

Recent computational-pathology work shows that routine H&E WSIs carry morphologic patterns linked to receptor status: Rawat et al. learned tissue "fingerprints" for ER/PR/HER2 prediction from H&E images [7], and Shamai et al. developed a clinically validated multi-institutional system for ER, PR, and HER2 prediction directly from H&E slides, showing morphology alone can support receptor inference at scale [8]; later work extended this via knowledge distillation and multiple-instance learning [9]. Across these studies performance is strongest for ER and more variable for PR and HER2, suggesting morphology alone is insufficient for the more molecularly heterogeneous targets, especially HER2. In parallel, transcriptomic studies established strong gene-expression associations with receptor status, and multigene assays demonstrated the clinical value of molecular profiling in breast cancer [5].

These findings motivate a unified framework exploiting morphologic, molecular, and clinical information jointly. Several gaps remain: most direct receptor predictors are unimodal and H&E-only, without transcriptomic or clinical context; multimodal pathomics frameworks developed for related tasks such as survival and subtyping [10] have not been systematically evaluated for joint ER/PR/HER2 prediction; and the per-modality contribution, the fusion benefit for the harder HER2 task, and cross-cohort generalizability remain insufficiently characterized.

To address these gaps, we propose an interpretable multimodal, multitask framework for joint ER/PR/HER2 prediction. Given clinical features, H&E WSI tile embeddings, and tumor gene-expression profiles from matched TCGA-BRCA cases [11], it learns modality-specific representations and fuses them by modality-level self-attention for patient-level prediction, with joint training across the three tasks to capture shared structure while preserving task-specific discrimi-

nation. The trained model is then evaluated on an independent CPTAC-BRCA cohort under strict external validation.

Our contributions are as follows:

1. We propose a multimodal, multitask deep learning framework that integrates clinical data, histopathology WSIs, and gene expression profiles via modality-level self-attention for joint ER/PR/HER2 prediction.
2. We perform systematic ablation across modality combinations and validate on an external cohort, demonstrating the effectiveness of multimodal fusion and cross-dataset generalization.
3. We incorporate interpretable modeling by linking predictions to receptor-associated genes and highlighting discriminative regions in WSIs via attention mechanisms.

Overall, the proposed framework is not intended to replace standard IHC/ISH testing. Rather, it is positioned as a decision-support approach that can complement conventional receptor assessment and may provide additional evidence in diagnostically challenging or heterogeneous cases.

2 Materials and Methods

2.1 Dataset

We used the TCGA-BRCA cohort [11] for development, assembling a patient-level multimodal cohort from three components: receptor-status labels, tumor RNA-seq gene expression, and WSIs. For external validation we used the independent CPTAC-BRCA [12] dataset, kept entirely separate from development and used only in a frozen setting (no retraining, tuning, feature reselection, or threshold adjustment). Receptor-status distributions for both cohorts are summarized in Table 1.

Table 1. Receptor-status distribution in the clinically annotated TCGA-BRCA development cohort and the CPTAC-BRCA external cohort

Dataset	Role	Patients	ER (+/−)	PR (+/−)	HER2 (+/−)
TCGA-BRCA	Development	937	725/212	626/311	191/746
CPTAC-BRCA	External	122	81/39	68/47	13/82

2.2 Data Processing

Clinical records, gene expression, and WSI embeddings were harmonized at the patient level, giving a final cohort of 937 patients with matched modalities. To prevent leakage, all distribution- or label-dependent preprocessing was fit within each training fold before application to the test fold.

Clinical Data. Clinical variables were cast to numeric or categorical form; missing numeric values were imputed with the training-fold median and missing categorical values assigned an `Unknown` category, followed by one-hot encoding.
Gene Expression. Each patient was represented by a tumor RNA-seq expression vector. We kept the 5,000 most variable genes, then applied mutual-information feature selection for ER, PR, and HER2 within each training fold; the selected genes formed the transcriptomic feature set, standardized using training statistics.
WSIs. Each H&E whole-slide image was represented as a set of precomputed tile embeddings from a pretrained pathology foundation model (UNI2-h) [13], each a 1,536-dimensional vector.

2.3 Model Architecture

As illustrated in Fig. 1, the framework contains three modality-specific branches followed by a shared multimodal fusion module. Clinical variables and gene-expression profiles are encoded by multilayer perceptrons, whereas WSI tile embeddings are processed by a Transformer-based multiple-instance learning encoder. The resulting modality embeddings are treated as separate tokens and fused using modality-level self-attention before being passed to a shared trunk and task-specific prediction heads for ER, PR, and HER2 classification.

For patient i, the complete multimodal input is denoted by

$$\mathcal{X}_i = \left\{ \mathbf{x}_i^{(c)}, \mathbf{x}_i^{(g)}, \mathbf{X}_i^{(w)} \right\}, \tag{1}$$

where $\mathbf{x}_i^{(c)} \in \mathbb{R}^{d_{\text{clin}}}$ denotes the clinical feature vector, $\mathbf{x}_i^{(g)} \in \mathbb{R}^{d_{\text{gene}}}$ denotes the gene-expression vector, and $\mathbf{X}_i^{(w)} \in \mathbb{R}^{K_i \times 1536}$ denotes the WSI tile-embedding matrix, in which K_i is the number of valid tiles retained for patient i and 1536 is the embedding dimension produced by the pretrained UNI2-h pathology foundation model [13].

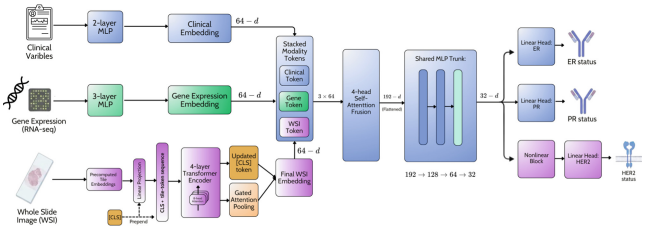


Fig. 1. Architecture of the proposed multimodal framework.

Clinical and Gene Expression Encoders. The clinical branch uses a two-layer MLP with dimensions $d_{\text{clin}} \to 128 \to d_f$, and the gene-expression branch uses a three-layer MLP with dimensions $d_{\text{gene}} \to 256 \to 128 \to d_f$. In both branches,

each hidden layer is followed by ReLU activation, batch normalization, and dropout. The modality embeddings are

$$e_i^{(c)} = \mathrm{MLP}_c(x_i^{(c)}), \qquad e_i^{(g)} = \mathrm{MLP}_g(x_i^{(g)}), \qquad e_i^{(c)}, e_i^{(g)} \in \mathbb{R}^{d_f}, \tag{2}$$

WSI Encoder. Each WSI is represented as a bag of precomputed tile embeddings $\mathbf{X}_i^{(w)} \in \mathbb{R}^{K_i \times 1536}$, where K_i is the number of valid image tiles retained for patient i. The value 1536 denotes the dimensionality of each tile embedding produced by the pretrained UNI2-h pathology foundation model, rather than a learned parameter of our model. To map this variable-length tile set into a shared latent space, each tile embedding is projected to a d_{wsi}-dimensional latent representation using a linear layer followed by LayerNorm, GELU activation, and dropout:

$$E_i = \phi_w\left(X_i^{(w)} W_p + \mathbf{1}_{K_i} b_p^\top\right), \qquad E_i \in \mathbb{R}^{K_i \times d_{\text{wsi}}}, \tag{3}$$

where $W_p \in \mathbb{R}^{1536 \times d_{\text{wsi}}}$ and $b_p \in \mathbb{R}^{d_{\text{wsi}}}$ are learnable projection parameters, $\mathbf{1}_{K_i}$ broadcasts the bias across tiles, and $\phi_w = \text{Dropout} \circ \text{GELU} \circ \text{LN}$.

To obtain a single representation for the whole slide, we prepend a learnable `[CLS]` token $c_{\text{cls}} \in \mathbb{R}^{1 \times d_{wsi}}$ to the projected tile-token sequence:

$$H_i^{(0)} = [c_{\text{cls}}; E_i] \in \mathbb{R}^{(K_i+1) \times d_{wsi}}. \tag{4}$$

The resulting sequence is then processed by a 4-layer pre-norm Transformer encoder with 8 attention heads:

$$H_i^{(\ell)} = \text{TransformerBlock}^{(\ell)}\left(H_i^{(\ell-1)}\right), \qquad \ell = 1, \ldots, 4. \tag{5}$$

Through self-attention, the `[CLS]` token aggregates information from all tile tokens, and its final hidden state is used as the slide-level embedding for downstream fusion and prediction.

Let $H_{i,k}^{(4)} \in \mathbb{R}^{d_{wsi}}$ denote the k-th row of $H_i^{(4)}$, and define

$$z_{i,\text{cls}} = H_{i,0}^{(4)}, \qquad h_{i,k} = H_{i,k}^{(4)}, \qquad k = 1, \ldots, K_i. \tag{6}$$

Since individual WSI tiles can have varying importance for predicting receptor status, we apply gated-attention pooling to the tile tokens, using a gating mechanism inspired by ABMIL [14]. The gated-attention pooled slide representation is

$$z_{i,\text{attn}} = \sum_{k=1}^{K_i} a_{i,k} h_{i,k}, \tag{7}$$

with attention weights

$$a_{i,k} = \frac{\exp(s_{i,k})}{\sum_{j=1}^{K_i} \exp(s_{i,j})}, \qquad s_{i,k} = w^\top(\tanh(V h_{i,k}) \odot \sigma(U h_{i,k})), \tag{8}$$

where $U, V \in \mathbb{R}^{(d_{wsi}/2) \times d_{wsi}}$ are learned projections, $w \in \mathbb{R}^{d_{wsi}/2}$ produces scalar attention scores, $\sigma(\cdot)$ denotes the sigmoid function, and $\odot$ denotes element-wise

multiplication. The final WSI embedding is obtained by concatenating the two readouts and projecting them into the shared fusion space:

$$e_i^{(w)} = f_w([z_{i,\mathrm{cls}}; z_{i,\mathrm{attn}}]) \in \mathbb{R}^{d_f}. \tag{9}$$

Modality-Level Self-attention Fusion. Considering that each modality contributes distinct information, the three modality-specific embeddings are concatenated and treated as separate modality tokens.

$$S_i = \begin{bmatrix} (e_i^{(c)})^\top \\ (e_i^{(g)})^\top \\ (e_i^{(w)})^\top \end{bmatrix} \in \mathbb{R}^{3\times d_f}, \tag{10}$$

and fused by a 4-head self-attention block followed by a residual connection and layer normalization:

$$T_i = \mathrm{LN}\big(\mathrm{MHA}(S_i, S_i, S_i) + S_i\big) \in \mathbb{R}^{3\times d_f}. \tag{11}$$

The attended tokens are then flattened into a fused representation

$$f_i = \mathrm{vec}(T_i) \in \mathbb{R}^{3\times d_f}. \tag{12}$$

Shared Trunk and Prediction Heads. The fused representation is passed through a shared multilayer trunk to learn a compact patient-level representation. This trunk progressively reduces the feature dimension and applies ReLU activation, batch normalization, and dropout after each layer. The shared representation is then used by task-specific prediction heads: ER and PR are predicted using linear heads, whereas HER2 is predicted using a slightly deeper nonlinear head to better capture its more complex and imbalanced decision boundary. Final receptor probabilities are obtained using sigmoid activation.

$$\ell_i^{(\mathrm{ER})} = W_{\mathrm{ER}} r_i + b_{\mathrm{ER}}, \qquad \ell_i^{(\mathrm{PR})} = W_{\mathrm{PR}} r_i + b_{\mathrm{PR}}, \tag{13}$$

$$\ell_i^{(\mathrm{HER2})} = W_2^{(h)} \phi_h\Big(W_1^{(h)} r_i + b_1^{(h)}\Big) + b_2^{(h)}, \tag{14}$$

2.4 Training Objective

Let $\mathcal{T} = \{\mathrm{ER}, \mathrm{PR}, \mathrm{HER2}\}$. For each task $k \in \mathcal{T}$, let $y_k \in \{0, 1\}$ denote the ground-truth label and $p_k \in (0, 1)$ the predicted probability of the positive class.

For ER and PR, we use focal loss:

$$\mathcal{L}^{(k)} = -\alpha_{y_k}^{(k)}(1 - q_k)^\gamma \log q_k, \qquad q_k = y_k p_k + (1 - y_k)(1 - p_k), \quad k \in \{\mathrm{ER}, \mathrm{PR}\}, \tag{15}$$

where $\alpha_{y_k}^{(k)}$ is the class-balancing weight.

For HER2, we use asymmetric loss to emphasize rare positive cases:

$$\mathcal{L}^{(\mathrm{HER2})} = \begin{cases} -(1 - p_{\mathrm{HER2}})^{\gamma^+} \log p_{\mathrm{HER2}}, & y_{\mathrm{HER2}} = 1, \\ -(p_{\mathrm{HER2},m})^{\gamma^-} \log(1 - p_{\mathrm{HER2},m}), & y_{\mathrm{HER2}} = 0, \end{cases} \tag{16}$$

where γ^+ and γ^- are the focusing parameters for positive and negative classes, and $p_{\mathrm{HER2},m} = \max(p_{\mathrm{HER2}} - m, 0)$ is the shifted negative-class probability with margin $m \in [0, 1)$, which suppresses the gradient contribution of easy negatives (samples already assigned a low predicted probability of the positive class).

The task losses are combined using uncertainty-based multitask weighting:

$$\mathcal{L}_{\mathrm{total}} = \sum_{k \in \mathcal{T}} \left(\frac{1}{2\sigma_k^2} \mathcal{L}^{(k)} + \log \sigma_k \right), \tag{17}$$

where $\sigma_k > 0$ is a learnable task-specific uncertainty parameter, optimized jointly with the model via $\sigma_k = \exp(\rho_k)$, $\rho_k \in \mathbb{R}$.

3 Experiments and Results

3.1 Experimental Setup

We evaluated the proposed framework on the TCGA-BRCA dataset using patient-level stratified five-fold cross-validation. In each run, one fold was for evaluation and the remaining folds were used for training. All data-dependent preprocessing steps, including missing-value imputation, feature standardization, variable-gene filtering, and mutual-information-based gene selection, were fitted only on the training split and then applied to the test split to prevent information leakage. The same model architecture and hyperparameter configuration were used across all folds. Performance is reported as the mean and standard deviation over the five test folds, with AUC and AUPRC used as the primary metrics because receptor-status labels, particularly HER2, are class-imbalanced. The training configuration is summarized in Table 2.

Table 2. Experimental protocol and training hyperparameters.

Component	Configuration
Training	Batch size 32; 200 epochs; dropout 0.2; weight decay 8×10^{-5}
Learning rate	3×10^{-4} for non-WSI parameters; 4×10^{-5} for the WSI branch
WSI encoder	$d_{\mathrm{wsi}} = 512$; 8 attention heads; 4 Transformer layers; maximum 512 tiles per WSI
Fusion dimension	$d_f = 64$
ER/PR objective	$\gamma = 2$
HER2 objective	$\gamma^+ = 1$ and $\gamma^- = 3.5$

3.2 Prediction Results

The proposed multimodal framework achieved strong overall performance for receptor-status prediction, with the best results observed for ER, followed by PR, while HER2 remained the most challenging task. As shown in Fig. 2, the

model reached AUC values of 0.954, 0.868, and 0.848 for ER, PR, and HER2, respectively, with similarly strong AUPRC for ER and PR and a lower but still competitive AUPRC for HER2.

As shown in Fig. 3, our model achieves the best ER (0.954) and HER2 (0.848) AUCs, improving by 2.1 and 3.0 percentage points over the strongest H&E baselines (Zhang Q. et al. [9] and Pignet et al. [15], respectively); for PR it reaches 0.868, marginally below Zhang Q.'s 0.887. On mean AUC, our model attains 0.890, exceeding all H&E-only baselines [9,15–18]. These results indicate that our approach delivers consistently strong multimodal performance, particularly on ER and HER2, while PR remains a task where high-performing pathology-only models are still highly competitive.

Table 3. Predictive performance on TCGA-BRCA and CPTAC-BRCA across ER, PR, and HER2 classification tasks.

Metric	Dataset	ER	PR	HER2	Macro avg.
AUC–ROC	TCGA-BRCA	0.954 ± 0.027	0.868 ± 0.026	0.848 ± 0.026	0.890
	CPTAC-BRCA	0.961 ± 0.018	0.926 ± 0.028	0.907 ± 0.051	0.931
AUPRC	TCGA-BRCA	0.983 ± 0.010	0.895 ± 0.036	0.668 ± 0.038	0.849
	CPTAC-BRCA	0.988 ± 0.006	0.959 ± 0.018	0.796 ± 0.104	0.914
F1-score	TCGA-BRCA	0.966 ± 0.011	0.903 ± 0.012	0.613 ± 0.046	0.827
	CPTAC-BRCA	0.938 ± 0.022	0.877 ± 0.033	0.581 ± 0.105	0.799
Accuracy	TCGA-BRCA	0.947 ± 0.017	0.862 ± 0.017	0.804 ± 0.036	0.871
	CPTAC-BRCA	0.911 ± 0.030	0.844 ± 0.038	0.856 ± 0.035	0.870

3.3 External Validation

We further evaluated cross-cohort generalizability on the independent CPTAC-BRCA cohort under a strict external validation setting [12,19]. After training on TCGA-BRCA, the complete pipeline was frozen and applied to CPTAC-BRCA without retraining, hyperparameter tuning, feature reselection, or threshold adjustment. As summarized in Table 3, the model maintained strong external performance across ER, PR, and HER2, with macro-average AUC–ROC and AUPRC values of 0.931 and 0.914, respectively. Performance was strongest for ER and PR, whereas HER2 remained the most challenging endpoint, as reflected by its lower AUPRC and F1-score.

3.4 Ablation Study

Across all seven modality combinations (Fig. 4), gene expression dominates the unimodal models (mean AUC 0.845, vs. 0.793 for WSI and 0.548 for clinical alone). Two-modality variants achieve the best ER (WSI + Gene, 0.958) and PR

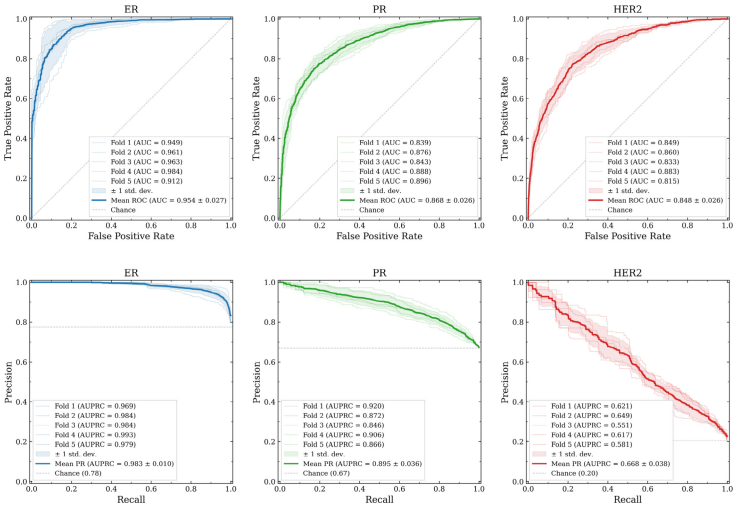


Fig. 2. Discriminative performance of the proposed multimodal model on TCGA-BRCA across five-fold cross-validation.

(Gene + Clin, 0.892), but only the full trimodal model attains the best HER2 (0.848) and the highest mean AUC (0.890), confirming that fusion benefits HER2 most.

3.5 Interpretability

We assessed interpretability by ranking transcriptomic features using mutual information with receptor status. Canonical receptor-associated genes were consistently among the most informative features, including *ESR1*, *GATA3*, and *AGR3* for ER; *PGR*, *ESR1*, and *CA12* for PR; and *ERBB2*, *GRB7*, and *TCAP* for HER2. Expression separation was clearer for ER and PR than for HER2, consistent with the corresponding prediction performance. Overall, these findings support the biological plausibility of the learned multimodal representation, although the analysis remains descriptive rather than causal.

We further examined slide-level attention maps to identify the histomorphologic regions that supported patient-level predictions. In a representative case (TCGA-3C-AALI), a 50-year-old woman with right-sided invasive ductal carcinoma, stage IIB (pT2N1aM0), the model correctly predicted ER-positive, PR-positive, and HER2-positive status (Fig. 5). The attention heatmap concentrated on tumor-rich regions while assigning relatively low weights to adipose and benign stromal areas. The most highly attended tiles contained densely packed malignant epithelial cells with nuclear hyperchromasia, pleomorphism, and loss of polarity, consistent with invasive ductal carcinoma. The attention distribution was spatially focused rather than diffuse: the top 100 tiles accounted for 12.2% of the total attention mass, with an entropy of 6.92 nats. Together, the gene-

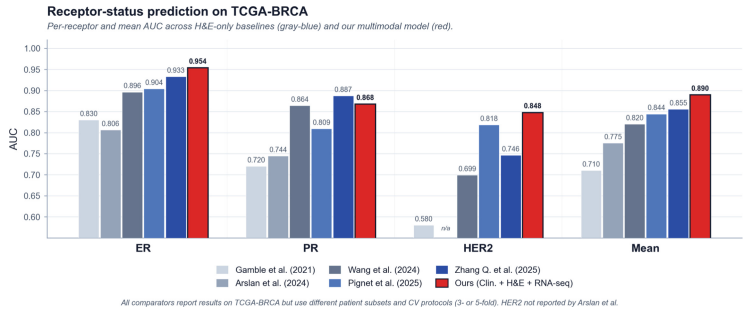


Fig. 3. Per-receptor and mean AUC for ER, PR, and HER2 prediction on TCGA-BRCA. Gray-blue bars denote H&E-only baselines (light to dark in order of mean AUC: Gamble [17], Arslan [16], Wang [18], Pignet [15], Zhang Q. [9]); the red bar is our multimodal model. All comparators report results on TCGA-BRCA but use different patient subsets and CV protocols; HER2 is not reported by Arslan et al. (Color figure online)

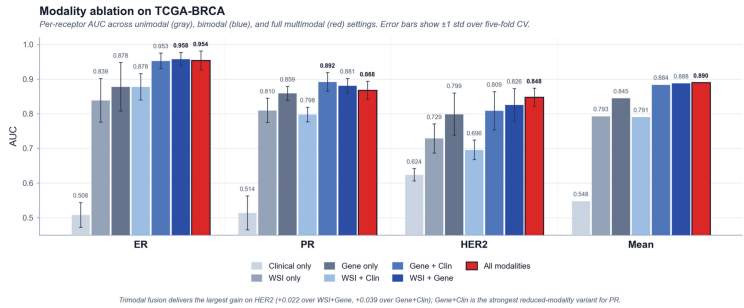


Fig. 4. Modality ablation on TCGA-BRCA. Per-receptor AUC across seven settings (3 unimodal in gray, 3 bimodal in blue, the full trimodal model in red). Error bars show ±1 std across five-fold cross-validation. (Color figure online)

ranking and attention analyses provide evidence that the model uses biologically and morphologically coherent signals for receptor-status prediction.

4 Discussion

4.1 Clinical and Methodological Implications

Receptor-status prediction benefits from combining complementary modalities rather than any single one: the multimodal model was more consistent across ER, PR, and HER2, with the clearest gain for HER2, indicating that morphology alone is insufficient for the more heterogeneous targets while molecular and clinical context strengthen patient-level inference. The ablation ranks transcriptomic features as the strongest unimodal signal, WSI as complementary morphologic evidence, and clinical variables as modest context, and the interpretability

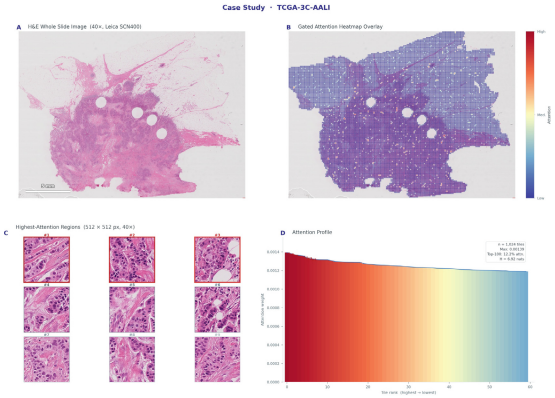


Fig. 5. Representative case showing the whole-slide image, tile-level attention heatmap, top-attended regions, and the attention scale.

analyses are biologically coherent (receptor-associated genes ranked most informative; attention maps concentrated on tumor-rich regions). We emphasize that the individual components are established; the contribution is their integrative design for joint, class-imbalance-aware ER/PR/HER2 prediction, together with the systematic seven-way modality analysis and frozen external validation on the independent CPTAC-BRCA cohort, whose consistent trimodal gain over the gene-only model indicates that fusion adds complementary signal rather than reproducing a molecular surrogate of the receptor labels.

4.2 Limitations and Future Work

Several limitations should be acknowledged. First, the external cohort was modest in size and imbalanced, particularly for HER2-positive cases, so the external estimates should be interpreted as preliminary. Second, the framework relies on RNA-seq, which may limit immediate clinical deployment; as discussed above, part of the predictive signal reflects molecular surrogates of receptor status, so a key goal is to retain the fusion benefit with reduced, clinically practical inputs.

Future work should evaluate larger multi-institutional cohorts, test robustness across heterogeneous acquisition settings, and develop reduced, clinically practical gene panels to improve translational feasibility. Overall, these findings support interpretable multimodal fusion as a promising strategy for ER, PR, and HER2 assessment in breast cancer. Rather than replacing standard IHC/ISH, the proposed framework may serve as a decision-support tool that complements conventional receptor testing and may be particularly useful in challenging cases.

5 Conclusion

In this paper, we presented an interpretable multimodal, multitask framework integrating clinical variables, H&E whole-slide images, and RNA-seq gene expres-

sion for joint ER/PR/HER2 prediction, developed on TCGA-BRCA and evaluated on the external CPTAC-BRCA cohort. Multimodal fusion helps most for HER2, the most heterogeneous and clinically challenging target, where all unimodal approaches show clear limitations. Intended to complement rather than replace standard IHC/ISH testing, the framework's main translational next step is a reduced, clinically practical gene panel that retains the multimodal gains—particularly for HER2—without requiring full RNA-seq.

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